

BHARAT ORAON

Urban Planning | Geospatial Analytics | Public Technology

Vijayawada, India

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Profile

Urban technologist and geospatial systems developer with applied experience in **urban governance, transport analytics, civic-tech infrastructure, and public-sector digital systems**. Worked on government-supported mobility platforms, GeoAI systems, and spatial intelligence tools translating complex urban datasets into actionable planning insights. Builds scalable full-stack geospatial platforms using FastAPI, React, Next.js, PostgreSQL/PostGIS, and cloud-based spatial analytics workflows.

Education

Bachelor of Planning (B.Plan)

Aug 2022 – Jun 2026

School of Planning & Architecture, Vijayawada

CGPA: 7.16 | Key Areas: Urban Governance, Transport Planning, GIS & Spatial Analysis, Urban Utilities, Development Planning

Experience

Build for Bharat Fellow (GIS) — Bharat Digital

Jan 2026 – Jul 2026

Chennai Unified Metropolitan Transport Authority (CUMTA)

- One of 19 fellows selected from 10,000+ applicants for the second cohort of the Build for Bharat Fellowship focused on public digital infrastructure and governance technology.
- Developed multimodal transit intelligence systems integrating Chennai Metro, Suburban Rail, and 3,000+ MTC bus stops using GTFS, PostgreSQL/PostGIS, Redis, Docker, React, Node.js, and MapLibre GL for mobility analysis across one of India's largest urban transport networks.
- Engineered a RAPTOR-based journey planning engine with multimodal routing and 400m transfer logic, enabling accessibility analysis and real-time route computation across Chennai's integrated public transit ecosystem.
- Designed and developed C-MARKS, a civic-tech platform digitizing road cutting permissions in Chennai with end-to-end spatial workflows, JWT-based multi-role authentication, and Google Street View-enabled virtual inspection tools for urban road infrastructure coordination.
- Developed GIS-enabled transit analytics dashboards visualizing ridership, interchange activity, accessibility metrics, and multimodal network performance across Bus, Metro, and MRTS systems using React-Leaflet, OpenStreetMap, and Chart.js.
- Engineered automated ETL pipelines using Python and Apache Airflow to process GTFS feeds and ChennaiOne ridership APIs into PostGIS databases with privacy-aware k-anonymity handling for large-scale public mobility datasets.

GIS Integration Intern (MoHUA-Sponsored Project)

May 2025 – Jul 2025

City Future Lab (IIT Kharagpur), Kharagpur, West Bengal

- Developed a full-stack WebGIS system enabling CRUD operations for spatial datasets using Python (GeoPandas), PHP, PostgreSQL/PostGIS, and Leaflet, supporting multi-user access and streamlining updates to transport and land-use layers.
- Improved bike-share rebalancing and bus scheduling visualization tools to support operational planning decisions.

Research Intern | Urban Climate & GIS

May 2025 – Jul 2025

OpenCity.in (Oorvani Foundation), Bengaluru / Remote

- Conducted urban heat and climate vulnerability analysis using satellite imagery and spatial indicators.

- Processed and analyzed remote-sensing datasets using Google Earth Engine, Python, and JavaScript.
- Authored three peer-reviewed articles published on OpenCity.in on urban heat, land-use change, and climate impacts in Indian cities.

Urban Planning Intern

May 2024 – Jul 2024

Patna Municipal Corporation, Patna, Bihar

- Supported planning and regulatory functions within a large Urban Local Body.
- Worked on building bye-laws, development regulations, and smart city initiatives.

Flagship Projects

Urban Equity Mapping Platform – Hyderabad

Live Project

- Built a full-stack GIS platform scoring all 145 GHMC wards on a composite Urban Equitability Index (UEI) using entropy weighting, K-Means clustering, Moran's I spatial autocorrelation, FastAPI, PostgreSQL/PostGIS, and Next.js.

Equicity AI – Urban Infrastructure Risk Assessment

Live Project

- Developed a GeoAI platform integrating Google Earth Engine satellite indices with Scikit-learn risk models to assess urban utility vulnerability through interactive spatial dashboards.

Karur Logistics Suitability Dashboard

Live Project

- Built a multi-criteria spatial suitability model for logistics hub site selection in Karur district using GeoPandas, Folium, and freight accessibility indicators.

Chennai Transfer Friction Index – Thesis Research

Live Project

- Quantified and mapped transfer friction across Chennai's multimodal transit network using pedestrian distance, interchange complexity, signage, and modal connectivity indicators.

Leadership & Activities

Public Relations Secretary

Feb 2025 – Mar 2026

Student Council, SPA Vijayawada

- Led institutional communication and outreach for the student body.
- Organized and hosted a National-level Inter-SPA Meet (2025).
- Overall Coordinator, Club-iO (Technical Club), SPA Vijayawada

Skills

Geospatial & Remote Sensing: QGIS, ArcGIS Pro, Landsat, Sentinel-2, Google Earth Engine (GEE), CARTO

Programming & Data Science: Python, Pandas, GeoPandas, NumPy, Scikit-learn, SQL, Apache Airflow

Web Development: React, Next.js, Node.js, Express.js, FastAPI, Streamlit

Web Mapping & Visualization: Mapbox GL JS, MapLibre GL, Leaflet.js, Kepler.gl, D3.js, Turf.js

AI & Spatial Systems: GeoAI Pipelines, Agentic Systems, Spatial Analytics

Databases & Infrastructure: PostgreSQL/PostGIS, Spatial SQL, Redis, REST APIs, Docker

Urban Planning & Policy: Governance, Land Use Planning, Transport Planning, Housing, Municipal Finance

Certifications

Summer Analytics – IIT Guwahati | Urban Land-Use and Transportation Planning – IIT Kharagpur (NPTEL) | Computer Science for Web Programming – Harvard University